Quantum Sensing and Atomic Physics Experimentation Kit

An experimental platform for atomic physics, spectroscopy and magnetometry.

The Quantum Sensing Experimentation Kit combines an External Cavity Diode Laser (ECDL) with a rubidium vapour system to provide hands-on exploration of spectroscopy and quantum magnetometry. It supports a progressive set of experiments covering ECDL handling and operation, laser stabilisation, rubidium hyperfine structure, sub-Doppler spectroscopy, polarisation-based measurements, magnetic-field-induced atomic level shifts, optical rotation and quantum magnetometry.

From textbook atomic physics to a measurement students can trust
No optical table, no dark room, no laser-safety officer for routine use.

By integrating ECDL frequency locking to an atomic reference with magnetic field sensing using the same atomic system, the kit gives students practical experience in complete atomic-system-based quantum sensing — the same measurement chain used in atomic clocks, magnetometers and inertial navigation.

Key Features

- Compact and user-friendly setup
- Easy to assemble and operate
- Long-lasting, alignment-free operation
- Intuitive software interface
- Integrated compact and stable laser system
- Integrated laser frequency stabilisation module
- Controlled and arbitrary magnetic field measurement
- Supports progression from basic optics to quantum sensing

Technical Specifications

Parameter	Unit	Min.	Max.	Typical
Operating Wavelength	nm	–	–	794.9 / 780.2
Optical Power	mW	–	–	30
Magnetic Field Source	μT	10	15000	10000
Magnetic Field Resolution	μT	–	–	10
Warm-up Time	mins	–	–	30
Continuous Adjustment-Free Duration	hrs	–	–	48
Weight	kg	–	–	15
Power Consumption	W	–	–	120
Ambient Temperature	°C	10	45	20
Dimensions	mm^3	–	–	415 × 480 × 110

What's Included

- The experimental kit — housing an ECDL, Rb vapour cell, optical components and photodetectors, with control and interfacing electronics
- Power, USB and LAN cables
- A pair of BNC cables

Experiments Included

Twelve experiments, progressing from foundational optics to research-level quantum sensing.

Exp. no	Experiment Name	Objective / Description
1	**Basics of Photodetection**	To understand detection of photons using APDs, crucial for measurement, spectroscopy and magnetometry
2	**Basics of Polarisation Optics**	To understand the basic principle of polarisation beam splitters, quarter- and half-wave plates and polarisers
3	**Basics of Interferometry**	To understand the basics of interferometry using a Mach–Zehnder interferometer as a tool for length metrology and refractive index measurement
4	**Zeeman Effect**	To understand the splitting of atomic energy levels and the resulting spectral line shifts caused by interaction with an external magnetic field
5	**Basics of ECDL**	To understand external cavity diode laser operation, basic handling and alignment, useful in all quantum optics experiments
6	**Doppler Spectroscopy in Atomic System**	To understand Doppler features in atomic spectroscopy

Exp. no	Experiment Name	Objective / Description
7	**Saturated Absorption Spectroscopy**	To understand the sub-Doppler technique using a pump-probe setup (hole burning) to resolve narrow atomic transitions
8	**Polarisation Spectroscopy**	To understand modulation-free, sub-Doppler spectroscopy using a pump–probe setup to resolve narrow atomic transitions
9	**DAVLL**	To understand modulation-free laser frequency stabilisation using Zeeman-induced polarisation-dependent absorption, without requiring a pump beam
10	**Magneto Optical Rotation (MOR)**	To understand rotation of laser beam polarisation due to an external magnetic field, used in magnetometry
11	**Magnetometry**	Measurement of magnetic fields using magneto-optical rotation in atoms
12	**Locking of ECDL**	Lock the frequency of the ECDL using a spectroscopic technique, useful in all quantum sensing experiments

Experiments 1 to 4 are foundational atomic and optical techniques suited to undergraduate teaching laboratories. Experiments 5 to 12 — spectroscopy techniques, magnetometry and laser frequency stabilisation — are genuine postgraduate and research-level tasks.

Learning Outcomes

Students will learn:

- Basics of photodetection
- Basics of polarisation optics
- ECDL handling and operation
- The Zeeman effect
- The full family of Doppler and sub-Doppler atomic spectroscopy techniques
- Interferometry and precision optical alignment
- Magneto-optical rotation and quantum magnetometry
- Laser frequency stabilisation using atomic spectroscopy

Skills Gained

Students will gain:

- Precision optics handling and alignment
- Interferometry
- Atomic spectroscopy
- Laser frequency stabilisation
- Quantum sensing

Industry Mapping

- Photonics R&D and optical instrumentation manufacturing
- Atomic clock and frequency-standards laboratories
- Metrology instrumentation, navigation and inertial sensing, geophysical exploration
- Defence sensing and secure defence communications
- Quantum sensor manufacturing

Atomic magnetometry and precision spectroscopy provide the foundation for applications in defence, navigation and geophysical exploration. Atomic magnetometers can detect weak magnetic signatures for applications such as submarine detection, enable GPS-independent navigation when integrated with inertial systems, and support mineral exploration and subsurface mapping when combined with gravimetric and magnetic surveying.

Curriculum Mapping

How the Kit Enables the Key Learning Outcomes	AICTE Course Title (Course Code)
Foundational experimental techniques in optics, data acquisition and processing	Basic Laboratory Course for Quantum Technologies (QT04)
Basics of photodetection and polarisation optics, interferometry and quantum optics	Introduction to Quantum Communication (QT06, QTM 11) · Quantum Optics (QT11) · Quantum Technology Lab 1 & 2 (QTM 06, QTM 12)
The basics of quantum measurement, quantum sensing and magnetometry	Introduction to Quantum Sensing (QT07, QTM 08)

QTXX refers to the AICTE Minor Degree for UG programmes in Quantum Technologies; QTMXX refers to the AICTE M.Tech Programme in Quantum Technologies.

Pre-requisites

- Basics of optics
- Basics of atomic physics
- Basics of interferometry and spectroscopy

Laboratory Requirements

- Clean and controlled laboratory environment — ambient temperature 18–30 °C, humidity 40–60%
- Bench space of 800 × 600 mm. No optical table required
- Basic laboratory equipment — oscilloscope, power meter and similar
- One earthed 230 V, 5 A socket within 2 m
- No dark room needed; avoid direct sunlight on the device
- Keep 2 m clear of large ferromagnetic objects and motors
- Network: one LAN port or Wi-Fi for data export (optional)
- A Windows 11 PC or laptop to control and interface with the device

⚠ Laser safety: handle the laser beam with care. Always follow appropriate laser safety practices and use suitable protective eyewear when required.

These are our recommended laboratory requirements. Please contact us if your laboratory configuration differs and we will work it out with you.

Frequently Asked Questions

What's included in the kit — cables, mounts, software licence, manuals?

The kit includes the Quantum Sensing and Atomic Physics experimental setup, power cord, USB and LAN cables, and a pair of BNC cables. The experimental setup houses an ECDL, Rb vapour cell, optical components and photodetectors, along with the control and interfacing electronics.

Is the kit suitable for UG, PG or PhD-level work?

Yes, across the range. Experiments 1 to 4 are foundational atomic and optical techniques and interferometry, suited to undergraduate teaching laboratories. Experiments 5 to 12: spectroscopy techniques, magnetometry and laser frequency stabilisation — are genuine postgraduate and research-level tasks.

Can experiments be customised or extended by faculty?

Yes. The experiments can be reprogrammed for other atomic-physics work, including alternative laser locking techniques and advanced magnetometry.

What safety precautions are required for magnetic field experiments?

The kit includes a magnetic coil. Keep metallic objects, including metal wearables, away from the experimental setup, as they may interfere with measurements and potentially damage the equipment.

What computer, operating system or software is needed to operate the kit?

The kit features an intuitive GUI to control the laser and other parameters. The software runs on a Windows PC, which is not included.

General

What laser safety precautions are required?

The lasers fall under the IEC 60825-1 Class 3B band, spanning 5 mW to 500 mW. This requires laser safety eyewear, a controlled-access laboratory area, and a beam enclosure where the beam is exposed. Exercise appropriate precautions when handling laser beams.

Is on-site installation and faculty or student training provided?

Yes, on-site installation and faculty and student training are provided.